

INTERNAL TECHNICAL REPORT | 2026-07-16

TWIRL repository audit and near-term execution plan

A reproducibility, production-safety, and scientific-readiness review of the standalone survey pipeline.

Abstract

We audited TWIRL as an executable survey pipeline, emphasizing reproducibility, production safety, validation semantics, artifact hygiene, and alignment between the Stage 1-5 plan and the code. The repository has a credible pilot foundation but is not survey-complete. The nearest defensible milestone is an enrichment-ready S56 periodic path: finish the bounded Tier-1 evidence, freeze a confident candidate/label set, and retrain teacher v1 while S59-S63 production continues independently. Dip search, multi-sector merging, false-alarm calibration, frozen recovery, and the release manifest remain mandatory before a survey-wide science product, but need not delay this bounded milestone.

Audit scope

Dimension	Evidence reviewed
Pipeline	Stage 1-5 implementation, interfaces, and acceptance gates
Validation	Fast tests, docs, detection sample, parsers, links, and syntax
Operations	Git branches/worktrees, artifact boundaries, local data, and production safety
Science	Claim boundaries, benchmark evidence, recovery semantics, and holdout use

Headline findings

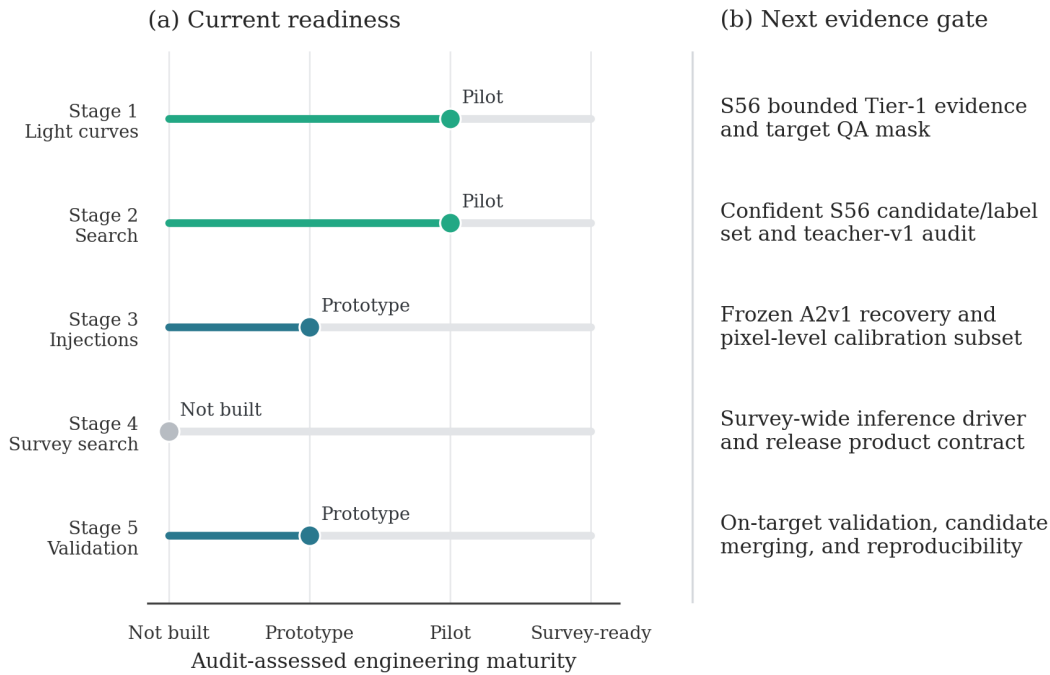
1. A2v1 generation, validation, compact export, and periodic BLS form a credible pilot path. 2. The fail-closed S56 Tier-1 code is implemented, but real authority artifacts must be pinned before enrichment. 3. Teacher-v2 and S57 labels remain exploratory. 4. Dip, multi-sector. and false-alarm work is deferred until the bounded periodic path is robust. not waived.

Bottom line: the next advance should be evidentiary rather than architectural.

Methods

The scan combined repository-wide inventory, Git/worktree inspection, parser and link checks, shell and Python syntax checks, the fast test suite, and review of the authoritative plan and protocols. The checkpoint contained 2,147 tracked files and 759.9 MiB; the fast suite completed with 312 passed and 3 skipped tests. We separated Tier-0 integrity from the bounded Tier-1 enrichment contract and from the still-future full-product science-release gate.

Pipeline readiness



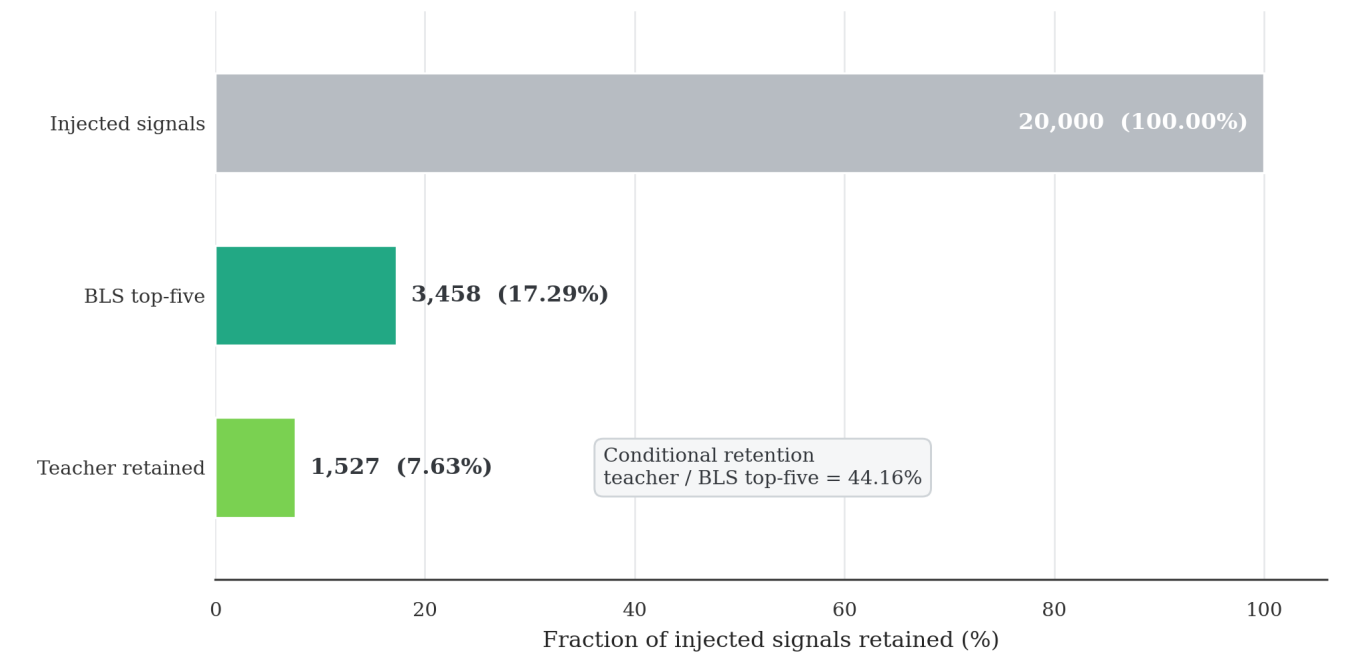
Readiness is an engineering audit judgment, not a scientific-performance score.

Figure 1. Audit-assessed engineering maturity and the next evidence gate for each pipeline stage. The scale describes implementation maturity, not scientific performance. Stage 1 and the periodic part of Stage 2 are at pilot maturity; Stage 3 and Stage 5 remain prototypes; Stage 4 is not yet built.

The strongest result is separation of production from analysis. The bounded Tier-1 implementation can qualify only the active S56 ADP pair and can never set science-ready. It intentionally cannot run until original-SPOC cadence evidence, exact canary-shard hashes, and an external WD 1856 product are pinned. Teacher-v2 and S57 labels should remain quarantined from the accepted baseline.

Results

Repository hygiene is acceptable at the source level but weak at the artifact-history boundary. At scan time the Git object store was approximately 14 GiB and dominated by historical generated outputs. The on-disk reports tree was approximately 5.5 GiB because it also contained large ignored or untracked review artifacts, while source, scripts, and tests together occupied only about 15 MiB. Nonportable symlinks, stale caches, large third-party reference bundles, and redundant rendered material belong outside versioned survey products. The local environment reported NumPy 2.2.6 although pyproject.toml pins NumPy below 2.0; tests passed, so this is environment drift rather than an observed failure.



Candidate-retention diagnostic only: this is not end-to-end survey completeness.

Figure 2. Candidate-retention diagnostic for the 20,000-signal teacher-v1 experiment. Top-five BLS retained 3,458 signals (17.29% of injections), and the teacher retained 1,527 (7.635% of injections; 44.16% conditional on top-five BLS recovery). This is not end-to-end survey completeness.

The attrition localizes losses, but it also shows why classifier iteration cannot substitute for baseline search and calibration. Reserve *end-to-end recovery* for a frozen chain that includes search, optional ranking, vetting, candidate merging, and the adopted calibration products. The present teacher result is a candidate-retention efficiency.

Period-radius candidate-retention surface

The aggregate retention counts localize pipeline attrition, while the canonical Teacher-v1 surface shows where the periodic search retains candidates across injected period, companion radius, and target brightness. It is included as diagnostic evidence, not as a final survey selection function.

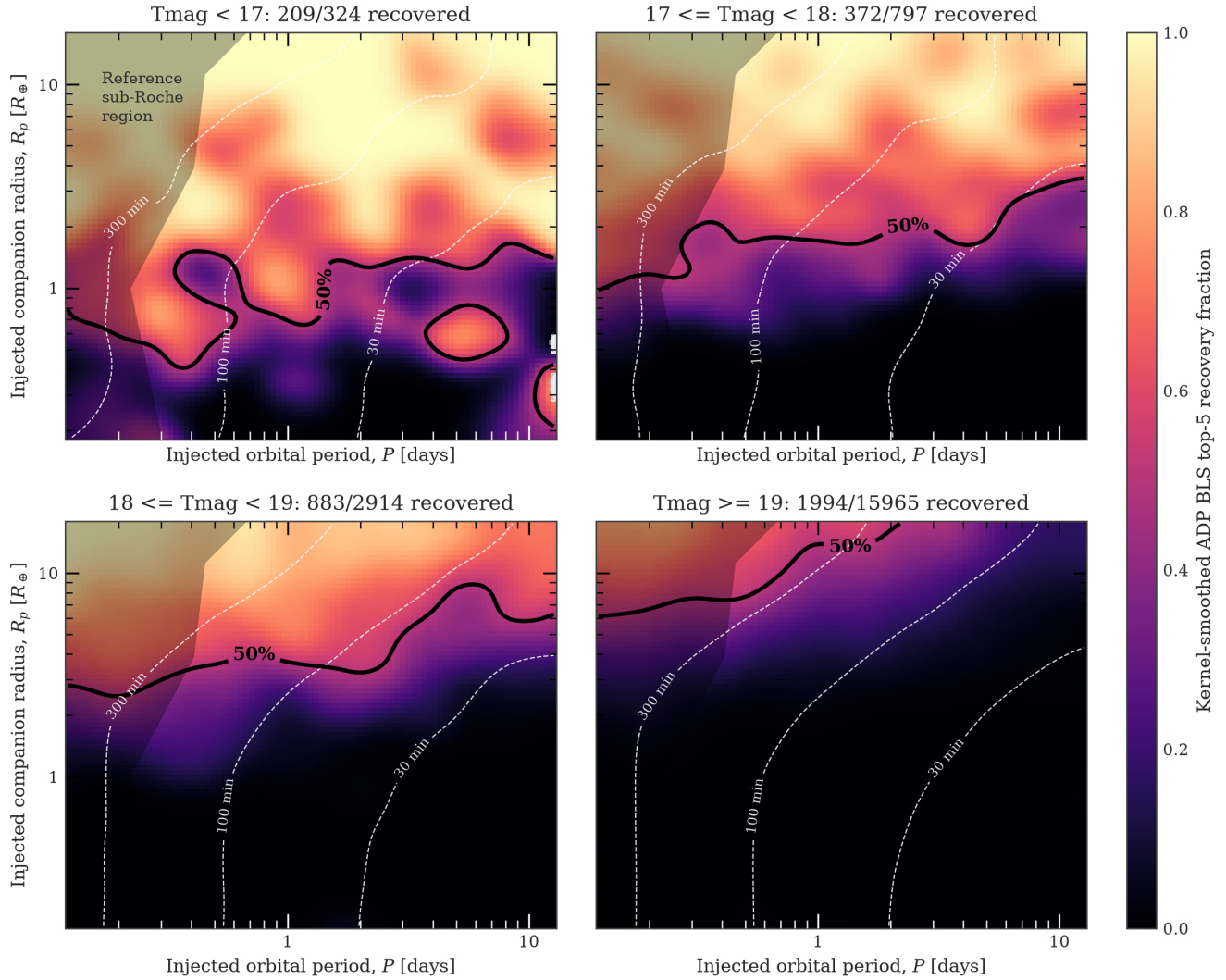


Figure 3. A2v1 Teacher-v1 BLS top-five candidate-retention surface for 20,000 injected signals, split into four T_{mag} strata. Fractions are kernel-smoothed; recovered and injected support counts are printed in every panel, and the bright bin is comparatively sparse. This is not end-to-end survey completeness and is not an occurrence-rate selection function.

A frozen survey analysis must propagate the same injections through the adopted vetter, candidate merger, and calibration chain before interpreting this surface as completeness.

Recommendations

1 Finish bounded S56 Tier-1	Build and pin the original-SPOC cadence reference, exact four canary shards, and external WD 1856 evidence; publish the TIC-level pass mask.
2 Freeze confident S56 labels	Apply the target mask, finish the periodic enrichment review, preserve both apertures, and merge only adjudicated candidates with provenance.
3 Harden teacher v1 around data	Keep the accepted model family; improve label manifests, TIC-grouped/source-separated tests, calibration, and bootstrap uncertainty.
4 Keep production independent	Let S59-S63 continue behind stop-on-failure gates. Do not wait for the queue before the S56 milestone; quarantine S57 labels and teacher v2.
5 Add deferred survey branches	After the periodic path is robust, add dip search, multi-sector merging, false-alarm calibration, and frozen-chain plus pixel-level recovery.
6 Freeze release and artifacts	Record the cutoff, sample, products, contracts, and hashes; version compact evidence and keep large rendered or local payloads outside Git.
7 Rebuild and automate	Rebuild from pyproject.toml before release validation and add PR CI for make test-fast and make check-docs after settling the lock strategy.

Execution order

Continue S59-S63 production in parallel; finish and pin bounded S56 Tier-1 evidence; freeze the confident periodic candidate/label set; retrain and audit teacher v1; then add dip, multi-sector, false-alarm, and frozen-recovery gates before survey-wide enrichment and science validation.

Conclusion

TWIRL has a testable pilot pipeline and a clear production contract. The best next advance is evidentiary: qualify S56 for bounded enrichment, turn it into a confident candidate/label set, and make teacher v1 auditable around the small real sample. Deferred search branches and full-chain calibration remain the boundary to a science-ready survey.

Reproducibility

Figures and this PDF are generated by scripts/build_repository_audit_report.py. The source report is repository_audit_report.md. The project plan and progress log remain the authoritative status records.